Alice P. Curtin

Canadian SKA Scientist at McGill University
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https://curtina.github.io

Research Interests

Fast radio bursts; Magnetars; Pulsars; Compact Object Mergers; Galactic Magnetic Field; Interstellar Medium; Radio Telescopes; Very Long Baseline Interferometry

Research Experience

Canadian Square Kilometre Array Scientist, McGill University — Fall 2025 - Present
Affiliated with the University of Amsterdam

Vanier Canada Graduate Fellow, McGill University — Fall 2022 - Fall 2025
Advisor: Victoria Kaspi

McPhee Fellow / Research Assistant, McGill University — Fall 2019 - Fall 2022
Advisor: Victoria Kaspi

Research Assistant, Carleton College — Jan. 2016 - June 2019
Supervisor: Joel Weisberg; Collaboration with Joanna Rankin

REU Research Assistant, University of Utah — June - Aug. 2018
Supervisor: David Kieda as a part of VERITAS & HAWC collaborations

Education

Doctor of Philosophy, Physics, McGill University — Sept. 2021 - Aug. 2025
Thesis: *Probing the Origins of FRBs using CHIME: High-energy Counterpart Searches and Burst Morphology*
Advisor: Victoria Kaspi
General GPA: 4.0

Master of Science, Physics, McGill University — Sept. 2019 - Sept. 2021
Thesis: *The Canadian Hydrogen Intensity Mapping Experiment Fast Radio Burst Project: Monitoring the Interference Environment and Studying the Bursting Behaviour of SGR 1935+2154*
Advisor: Victoria Kaspi
General GPA: 4.0

Bachelor of Arts, Physics & Astronomy, Carleton College — Sept. 2015 - June 2019
Thesis: *Jets in Active Galactic Nuclei*
Advisor: Joel Weisberg
General GPA: 3.75

Awards and Recognitions

Individual

Canadian Square Kilometre Array Scientist, $600k — 2025 - Present
Best Graduate Student Talk at Canadian Astronomical Society Meeting — 2023

Honorable Mention for EPO Poster at Canadian Astronomical Society Meeting 2023
McGill Department of Physics Community Building Award, $1000 2023
Vanier Canada Graduate Fellow, NSERC $150000 2022 - 2025
(Ranked #6 out of all NSERC Vanier candidates in Canada)
Distinction in Physics & Astronomy Bachelor Degree, Carleton College 2019
Distinction on Physics Thesis, Carleton College 2019
Honorable Mention for Research, University of Utah 2018
Mike Ewers Award, Minnesota Space Grant Consortium, Carleton College, $1000 2018
Minnesota Space Grant Consortium Award, Carleton College, $1000 2016

Team

Women and Girls in Astronomy Program 2026 Cohort, $2800 2026
(One of ten selected outreach initiatives in North America)
President's Prize for Public Engagement, as a part of the McGill Eclipse team 2025
Marcel Grossmann Award, as part of the CHIME/FRB team 2024
McGill Faculty of Science Funding for Science in Space Outreach Initiative, $5000 2023
2022 Brockhouse Canada Prize, as part of the CHIME team 2022
Lancelot M. Berkeley - New York Community Trust Prize, as part of the CHIME/FRB team 2022
McGill University McPhee Fellowship, $10000 2020
Governor General's Innovation Award, as part of the CHIME team 2020

Teaching & Mentorship Experience

Summary: 500+ hours of teaching experience; Formally mentored six undergraduate students, two MSc students

Courses

Curriculum designer & lead, Radio Astronomy Course, McGill Winter 2027
Curriculum designer & lead, Canadian Radio Astronomy Summer School Summer 2026
Co-organizer, Graduate-level Astrostatistics Reading Course, McGill University Fall 2023

Lectures

Canadian Radio Astronomy Summer School Summer 2026
Radio Astronomy Course, University of British Columbia - Okanagan, Winter 2026
Graduate-level Astrostatistics Reading Course, McGill University Fall 2023
Advanced E&M, Carleton College, Winter 2019

Teaching Assistantships

McGill University

Teaching Assistant, Topics in Classical Mechanics Winter 2021
Lab Designer, Facilitator, and Grader for Introductory Physics Course Summer - Fall 2020
Lab Assistant and Grader for Introductory Electricity and Magnetism Fall 2019
Lab Assistant and Grader for Introductory Mechanics Fall 2019

Carleton College

Grader for Math 341, Fourier Series and Boundary Value Problems Spring 2019

Lab Assistant for Physics 165, Electricity and Magnetism — Winter 2019
Problem Solving Facilitator for First and Second Year Physics — Winter 2019

Graduate Mentees

Lucas Kritz, MSc — Fall 2026 - Present
Melanie Szpigiell, MSc — Fall 2026 - Present
Sloane Sirota, *External thesis committee member* — Spring 2026 - Present

Undergraduate Mentees

Jasmine Boyd*, Co-supervised with M. Chamma & J. Hessels — Fall 2026 - Present
Karina Kaplina, Co-supervised with A. Cook & J. Hessels — Summer 2026 - Present
Audrey Bernier*, Co-supervised with J. Hessels — Summer 2025, Winter - Summer 2026
Melanie Szpigiell, Co-supervised with V. Kaspi — Summer 2025, 2026
Sloane Sirota, Co-supervised with V. Kaspi — Summer 2023 - Winter 2024
Sandhya Rotoo, Co-supervised with V. Kaspi — Summer 2022

Academic Leadership & Service

Summary: Co-chair for four conferences; Reviewer for three academic journals & one book; Served on multiple fellowship committees & instrumentation working groups

NSERC Fellowship (MSc, PhD) Review Committee (*Invited*) — Fall 2025, 2026
Co-chair, Canadian Astronomical Society Meeting 2027 (*Invited position*) — Winter 2026 - Present
CHIME/FRB Operations Co-Lead (*Invited position*) — Summer 2025 - Present
Co-convener of Counterparts Working Group, CHIME/FRB — Winter 2025 - Present
Square Kilometre Array Transient Science Working Group — Winter 2025 - Present
CHIME Radio Frequency Interference Committee (*Invited position*) — Fall 2024 - Present
Canadian Astronomical Society Sustainability Committee — Summer 2024 - Present
Reviewer: MNRAS†, ApJ‡, PASA§, Square Kilometre Array Book — Fall 2023 - Present
Co-founder & convener of FRB¶ Early Career Researcher Journal Club — Fall 2023 - Present
Chair of the first Canadian Square Kilometre Array Community Meeting — Summer 2026
Chair of the first Canadian Radio Astronomy Summer School — Summer 2026
LOC for Dynamic Radio Sky Conference — Summer 2026
Co-chair of Local Organizing Committee for FRB 2025 — Fall 2025
Co-chair of Scientific Organizing Committee for FRB 2025 — Fall 2025
Pipeline Expert & Admin, CHIME/FRB (*Invited position*) — Fall 2023 - Summer 2025
Convener of McGill Transient Discussion — Spring 2022 - Winter 2025

Community Leadership & Engagement

Summary: I have spent over 1500 hours on outreach and science communication.

Canada Excellence Research Chair EDI Committee, McGill University — Winter 2025 - Present

*Senior thesis
†Monthly Notices of the Royal Astronomical Society
‡Astrophysical Journal
§Publications of the Astronomical Society of Australia
¶Fast Radio Burst

Co-founder & Principal Member of Science in Space Outreach Initiative, McGill University, Dell Technologies	Spring 2022 - Present
McGill Physics Outreach Committee Member[‖]	Spring 2020 - Present
Co-organizer Graduate Outreach Session for Canadian Astronomical Society Meeting 2025 (*Invited*)	Spring 2025
Writer & Committee Member for Astrobites	Winter 2021 - Fall 2024
Judge & Selection Committee Member for CCUWiP* Conference (*Invited*)	Winter 2024
Mentorship Panelist for Graduate School (*Invited*), McGill University	Fall 2023
EDI Committee, McGill University	Summer 2020 - Spring 2021
Volunteer for Goodsell Observatory, Carleton College	Summer 2016 - Summer 2019
Physics Department Curriculum Committee, Carleton College	Fall 2018 - June 2019
Science Summer Educator, Berkshire Museum, Pittsfield, MA	Summer 2017

Publications

Summary: Six first-author refereed publications; four first-author non-refereed contributions; twelve publications as a lead co-author. Total: 60+ works; 2900 citations; h-index: 25.

Refereed First-Author Publications

1. **Curtin, A.P.** on behalf of the SKA Transients SWG (2026), The astrophysics of fast radio bursts, Chapter in Advancing Astrophysics with the SKA – II. doi:10.48550/arXiv.2606.27546
2. **Curtin** et al. (2026), CHIME/FRB Discovery and Localization of the Swift-observed FRB 20241228A, ApJ, 998, 97. doi:10.3847/1538-4357/ae2ea0
3. **Curtin** et al. (2025), Morphology of 35 Repeating Fast Radio Burst Sources at Microsecond Time Scales with CHIME/FRB, ApJ, 992, 206. doi:10.3847/1538-4357/adf844.
4. **Curtin**, Weisberg, & Rankin (2024), Determining the Magnetic Field in the Galactic Plane from New Arecibo Pulsar Faraday Rotation Measurements, ApJ, 975, 215, https://doi.org/10.3847/1538-4357/ad7b15.
5. **Curtin** et al. (2024), Constraining Near-simultaneous Radio Emission from Short Gamma-Ray Bursts Using CHIME/FRB, ApJ, 972, 125, https://doi.org/10.3847/1538-4357/ad5c65.
6. **Curtin** et al. (2023), Limits on Fast Radio Burst-like Counterparts to Gamma-Ray Bursts Using CHIME/FRB, ApJ, 954, 154, https://doi.org/10.3847/1538-4357/ace52f.

Non-Refereed Contributions

1. **Curtin, A.P.** et al. (2026), One Attempt at Building an Inclusive & Accessible Hybrid Astronomy Conference: FRB 2025, doi:10.48550/arXiv.2601.14357
2. **Curtin, A. P.** & Cook A.M. (2025), Fast Radio Bursts 2025, NatAs, 9, 1760. doi:10.1038/s41550-025-02741-1

[‖]Outreach initiative coordination and facilitation including public talks, school visits, and large-scale event coordination. I am the longest-standing member (out of all faculty, staff, students) on the McGill Physics Outreach Committee. Served as a paid graduate coordinator from 2021-2025.

*Canadian Conference for Undergraduate Women & Gender Minorities in Physics

3. **Curtin, A.P.** and CHIME/FRB Collaboration (2024), "CHIME/FRB Updated Position for Repeating Source FRB 20240316A," ATel 16780, https://www.astronomerstelegram.org/?read=16780.

4. **Curtin, A.P.** and CHIME/FRB Collaboration (2023), "Non-detection of radio emission from GRB 231115A with CHIME/FRB," ATel 16341, https://www.astronomerstelegram.org/?read=16341.

Co-authored Contributions: A full list of my contributions can be found through the Astrophysics Data System ↗.

Observing Proposals

Summary: PI for 302 hours of awarded time with the Very Long Baseline Array (VLBA), and co-PI on an additional 130 hours with the Green Bank Telescope (GBT), Very Large Array (VLA), and VLBA.

(Co-PI) *Assessing the Frequency-Dependent Nature of VLBI Calibrators for CHIME/FRB* — VLBA 2026
PI: Hewitt, Hours Acquired: 22

(Co-PI) *The First Radio Survey of Fast Radio Burst Host Environments* — VLA 2025
PI: Dong & Lazda, Hours Acquired: 60

(Co-PI) *Constraining the nebular model of fast radio bursts with three new localizations* — VLBA 2025
PI: Lazda, Hours Acquired: 25

(Co-PI) *Unveiling compact radio emission associated with a local universe FRB* — VLBA 2025
PI: Lazda, Hours Acquired: 8

(Co-PI) *Confirming Radio Detection of Magnetar SGR J0418+5729* — VLA 2025
PI: Sherman, Hours Acquired: 1.4

(Co-PI) *Searching for Radio Emission from the Four 'Low B-Field' Magnetars* — GBT 2025
PI: Sherman, Hours Acquired: 14.25

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA 2024
PI: Curtin, Hours Acquired: 80

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA 2023
PI: Curtin, Hours Acquired: 180

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA 2022
PI: Curtin, Hours Acquired: 42

Invited & Contributed Talks

Summary: 14 invited talks; 15 seminars or colloquia; 19 contributed or shorter institutional talks

Seminars & Colloquia

Bishop's University, Astrophysics seminar (*Invited*) — Sherbrooke, QC, Winter 2027
University of Minnesota, Physics colloquium (*Invited*) — Minneapolis, MN, 2026
Queen's University, Astrophysics seminar (*Invited*) — Kingston, ON, 2026
Cornell University, Astrophysics lunch — Ithaca, NY, 2026
John Abbott College, Seminar — Montreal, QC, 2026
Dominion Astrophysical Observatory, Science Tea Seminar — Victoria, BC, 2026

Dominion Radio Astrophysical Observatory, Seminar (*Invited*)	Penticton, BC, 2026
University of Wisconsin–Madison, Science Seminar Series	Online, 2026
University of Vermont, Physical Society Colloquium (*Invited*)	Burlington, VT, 2025
Marianopolis College, Seminar (*Invited*)	Montreal, QC, 2024
Institut d'Astrophysique Spatiale, Seminar (*Invited*)	Paris, France, 2024
CIERA, Northwestern University, Seminar	Evanston, IL, 2024
Caltech, Radio Seminar	Pasadena, CA, 2024
McGill University, Seminar (*Invited*)	Montreal, QC, 2023
Dominion Radio Astrophysical Observatory, Tech Talk (*Invited*)	Online, 2021

Conferences

Dynamic Radio Sky 2026 (*Invited*)	Montreal, QC, 2026
Canadian Astronomical Society Meeting, Education & Public Outreach talk	Montreal, QC, 2026
Canadian Astronomical Society Meeting, Science talk	Montreal, QC, 2026
Canadian Astronomical Society Meeting, SKA[†] lunch (*Invited*)	Montreal, QC, 2026
Canadian Astronomical Society Meeting, Graduate student day (*Invited*)	Montreal, QC, 2026
Canadian Square Kilometre Array Community Meeting	Online, 2026
CARTA[‡] Meeting (*Invited*)	Victoria, BC, 2026
Rutgers Transients Conference (*Invited*)	Online, 2025
Structure and Polarization in the Interstellar Medium	Online, 2024
Fast Radio Burst 2024	Thailand, 2024
Canadian Astronomical Society Meeting, Education & Public Outreach talk	Toronto, ON, 2024
Fast Radio Burst 2023	Online, 2023
Canadian Astronomical Society Meeting	Penticton, BC, 2023
Fast Radio Burst 2022	Busan, South Korea, 2022
Centre for Research in Astrophysics of Quebec, Annual Meeting	Quebec, QC, 2022
Radio Frequency Interference 2022, two contributions	Online, 2022

Shorter institutional talks

University of Manchester Transient Journal Club	Manchester, UK, 2026
University of Amsterdam Transient Journal Club	Amsterdam, NL, 2026
Rutgers University Astronomy Journal Club	Online, 2025
MIT Kavli Institute Tea Talk	Cambridge, MA, 2024
UC Berkeley Astronomy Lunch Talk	Berkeley, CA, 2024
Northwestern CIERA Observer's Group Meeting	Online, 2023
West Virginia University Astronomy Journal Club	Online, 2023

Science Communication Articles & Media

Summary: 12 science communication articles; three social media clips

[†]Square Kilometre Array
[‡]Cube Analysis and Rendering Tool for Astronomy

1. A. P. Curtin, "Some 'not so fast' fast radio bursts," Astrobites, November 2022.
2. A. P. Curtin, "An FRB way off in the distance," Astrobites, October 2022.
3. A. P. Curtin, "Have we found the origins of fast radio bursts?," Astrobites, September 2022.
4. A. P. Curtin, "Could some short and long gamma-ray bursts have the same parents?," Astrobites, May 2022.
5. A. P. Curtin, "You'll be a limbo star. How (s)low can you go?," Astrobites, February 2022.
6. A. P. Curtin, "Let's get building (some terrestrial planets)!," Astrobites, December 2021.
7. A. P. Curtin, "Another Mysterious Fast Radio Burst Detected… Are We One Step Closer to Discovering their Origins?," Astrobites, November 2021.
8. A. P. Curtin, "New Radio Source Towards the Center of our Galaxy – Say whaaaat," Astrobites, October 2021.
9. A. P. Curtin, "A Fast Radio Burst in a Rather Peculiar Location," Astrobites, August 2021.
10. A. P. Curtin, "If you had $100 million, how would you look for aliens?," Astrobites, May 2021.
11. A. P. Curtin, "FRBs are spiraling out of control," Astrobites, March 2021.
12. A. P. Curtin, "Three Little Outliers in a Sea of Planets, Stars, and Brown Dwarfs," Astrobites, February 2021.
13. A. P. Curtin, Instagram reel on NANOGrav gravitational wave detection, Astrobites, July 2023, 3000 views.
14. A. P. Curtin, Instagram reel on renewable energy in the South Pole, Astrobites, December 2023, 1500 views.
15. A. P. Curtin, Instagram reel on 7 eclipse facts, Astrobites, April 2024, 260,000 views.

Press Release

Summary: Six press releases that focus on my work

1. *Training Canada's next generation of radio astronomers* ↗
   McGill Reporter 2026
2. *Fast Radio Burst Conference hosted at McGill for the first time* ↗
   McGill Faculty of Science 2025
3. *First early-career astronomers selected as Canadian SKA Scientists* ↗
   National Research Council Canada 2025
4. *McGill outreach initiative introduces astronomy to girls in grades 5 and 6* ↗
   McGill Reporter 2025
5. *Bringing physics and space to life in Montreal classrooms* ↗
   McGill Impact 2023
6. *Shedding light on the mysteries of the universe* ↗
   McGill Giving 2022